Ashutosh Tripathi

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RESEARCH INTERESTS

- Theoretical High Energy Physics
- Lattice Gauge Theory/QCD
- Matrix Models (IKKT, BFSS)
- Classical and Quantum simulations.
- Quantum Computing
- Artificial Neural Networks
- Holography and Quantum Chaos
- Langevin dynamics, GTM, Tensor Networks

RESEARCH SUMMARY

I am broadly interested in non-perturbative approaches to quantum field theory, lattice gauge theory, and matrix models motivated by string theory and quantum gravity. My work spans both continuum and lattice formulations, focusing on overcoming computational challenges such as the sign problem, topological freezing, and large-N scaling. I have explored a range of techniques—from Lefschetz thimbles and complexified path integrals to variational methods based on neural network quantum states. More recently, I am investigating how machine learning can enhance these methods by encoding physical symmetries and topological features, with the aim of enabling scalable simulations of strongly coupled systems. My long-term goal is to develop robust computational frameworks that bridge fundamental theory and modern AI to address open problems in high-energy and quantum many-body physics.

RESEARCH POSITIONS

◆ **Queen Mary University of London** — **June 2025**, London, UK
Position : Visitor
Department : School of Mathematical Sciences
Supervisor : Dr. Masanori Hanada

◆ **RIKEN** — **April 2025 - Sept 2025**, Saitama, Wako, Japan
Position : Student Trainee
Lab : Theoretical Quantum Physics Laboratory
Host : Prof. Franco Nori

◆ **High Energy Accelerator Research Organization (KEK)** — **November 2022 - Sept 2025**, Tsukuba, Japan
Position : Research Assistant

EDUCATION

◆ **Ph.D. in Theoretical Physics** — **October 2022 - today**
High Energy Accelerator Research Organization (KEK), Tsukuba, Japan
Registered at : The Graduate University for Advanced Studies (SOKENDAI), Hayama, Japan
PhD thesis title : Numerical Studies of Non-Perturbative Dynamics in IKKT and BFSS Matrix Models
Supervisor : Prof. Jun Nishimura.
Expected Date of Graduation: September 2025

◆ **Integrated Bachelor's and Master's (BS-MS) in Physics** — **August 2017 - May 2022**
Indian Institute of Science, Education and Research (IISER), Mohali, INDIA
MS-Thesis : Non-lattice simulations of supersymmetric Yang-Mills theories
Supervisor : Dr. Anosh Joseph
CPI : 9.0/10 [transcript]

PUBLICATIONS [1]

Total citations: 5, **h-index:** 1
iNSPIRE HEP

1. **Inequivalence between the Euclidean and Lorentzian versions of the type IIB matrix model from Lefschetz thimble calculations**
   Authors: Chien-Yu Chou, Jun Nishimura, **Ashutosh Tripathi**
   Published in : Phys. Rev. Lett. 134, 211601
   DOI : https://doi.org/10.1103/PhysRevLett.134.211601

ONGOING RESEARCH PROJECTS [2]

◆ **Monte Carlo studies of the IKKT matrix model using generalized Lefschetz thimble method**
Supervisor – Prof. Jun Nishimura (KEK)

1. Investigated both Euclidean and Lorentzian formulations of the type IIB (IKKT) matrix model—proposed as a non-perturbative definition of superstring theory—via first-principles Monte Carlo simulations.
2. Successfully implemented the generalized Lefschetz thimble method (GTM) to overcome the severe sign problem and demonstrated a clear inequivalence between the Euclidean and Lorentzian models at matrix size $N = 2$.
3. Developed a gauge-fixing prescription for the noncompact Lorentz symmetry, enabling a fully nonperturbative formulation compatible with complexified integration contours.
4. Ongoing work includes scaling the simulations to larger N and incorporating dynamical fermions to explore supersymmetry and emergent spacetime structure.

◆ **Variational Monte Carlo with Neural Network Quantum States for Yang-Mills Matrix Models**
Supervisor: Dr. Masanori Hanada (Queen Mary University of London, UK)

1. Implementing the variational Monte Carlo framework using neural autoregressive flow architectures to approximate ground-state wavefunctions of SU(N) Yang-Mills-type matrix models.
2. Benchmarked variational results against Lattice Monte Carlo simulations to verify accuracy and convergence.
3. Demonstrated that increasing neural network width improves precision; investigated finite-α effects and convergence trends.
4. Extended the method to larger matrix size N, higher-dimensional matrix models (3, 5, 9 matrices), and models with supersymmetric degrees of freedom.

◆ **Lefschetz Thimble Analysis of 2D U(1) Gauge Theory with θ-Term**
Supervisors: Prof. Jun Nishimura (KEK), Dr. Kohtaroh Miura (KEK)

1. Studying 2D compact U(1) lattice gauge theory with a topological θ-term, which induces a sign problem relevant to phenomena like instantons and axions.
2. Employed the generalized Lefschetz thimble method to tame the complex phase fluctuations and validate against the analytically solvable structure of the model.
3. Developed strategies to address topological freezing by introducing punctures on the toroidal lattice geometry.

TALKS DELIVERED

8. **Numerical studies of the IKKT matrix model with the gauge fixed Lorentz symmetry**
   Date - Dec 17th, 2024
   Place - RIKEN
   Webpage - 7th Intensive Lectures by Quantum Gravity Gatherings, 17-19th December, 2024 [Slides]

7. **Numerical studies of the type IIB matrix model with the gauge fixed Lorentz symmetry**
   Date - Dec 12th, 2024
   Place - KEK
   Webpage - KEK Theory Workshop 2024, 11-13th Dec, 2024 [Slides]

Authors are mostly listed in alphabetical order.
These projects are in progress and may result in future publications.

6. **Fermions, quantum gravity and holography in two dimensions**
   Date - Mar 25, 2024
   Place - KEK Theory Center (Journal Club Presentation) [Slides]

5. **Lefschetz-thimble analysis of the Lorentzian IKKT matrix model around its classical solutions**
   Date - Mar 20, 2024
   Place - Virtual
   Webpage - The Physical Society of Japan 2024 Spring Meeting (Virtual), 18-21st Mar, 2024 [Slides]

4. **Lefschetz-thimble analysis of the Lorentzian IKKT matrix model around its classical solutions**
   Date - Nov 29, 2023
   Place - KEK
   Webpage - KEK Theory Workshop 2023, 29th Nov - 1st Dec, 2023 [Slides]

3. **Lefschetz thimble analysis of the Lorentzian IKKT matrix model around saddle-point configurations**
   Date - Sept 19, 2023
   Place - Tohoku University
   Webpage - The Physical Society of Japan 2023 Annual Meeting (All divisions), 16th - 19th Sep, 2023 [Slides]

2. **Numerical Analysis of the Lorentzian type IIB matrix model using generalized Lefschetz thimble method**
   Date - Apr 27th, 2023
   Place - RIKEN
   Webpage - Spring Workshop on Quantum Gravity, 26th - 28th Apr, 2023 [Slides]

1. **Numerical Sign problem and Lefschetz Thimble method**
   Date - Mar 31st, 2023
   Place - KEK Theory Center (Journal Club Presentation) [Slides]

POSTERS PRESENTED

6. **Inequivalence between the Euclidean and Lorentzian versions of the type IIB matrix model from Lefschetz thimble calculations**
   Date - Mar 13th, 2025
   Place - Kyushu University Institute for Advanced Study
   Webpage - Kyushu IAS-iTHEMS conference: Non-perturbative methods in QFT, 10th-14th Mar, 2025 [Poster]

5. **Monte Carlo studies of IKKT matrix model using generalized Lefschetz thimble method**
   Date - Nov 12th, 2024
   Place - KEK Student Day 2024 [Poster]

4. **Lefschetz thimble analysis of the Lorentzian IKKT matrix model around its classical solutions**
   Date - Dec 6th, 2023
   Place - Yukawa Institute for Theoretical Physics, Kyoto University
   Webpage - The 18th Kavli Asian Winter School on Strings, Particles and Cosmology, 5th - 14th Dec, 2023 [Poster] [Gong Show]

3. **Numerical studies of the Lorentzian IKKT matrix model**
   Date - Nov 14th, 2023
   Place - KEK Student Day 2023 [Poster]

2. **The Sign Problem and Lefschetz Thimble Method**
   Date - Jan 9th, 2023
   Place - Institute for Basic Science, Daejeon, South Korea
   Webpage - The 17th Kavli Asian Winter School on Strings, Particles and Cosmology, 9th - 18th Jan, 2023 [Poster] [Gong Show]

1. **Non-lattice simulations of Supersymmetric quantum mechanics model**
   Date - Dec 19th, 2021
   Place - Master's Thesis Defense [Poster]

WORKSHOPS/CONFERENCES ATTENDED

16. Kyushu IAS-iTHEMS conference: Non-perturbative methods in QFT, 10th - 14th Mar 2025, Kyushu University, Institute for Advanced Study, Fukuoka, Japan.

15. 7th Intensive Lectures by Quantum Gravity Gatherings, 17th - 19th Dec 2024, RIKEN, Wako, Japan. [ID]

14. KEK Theory Workshop 2024, KEK Theory Center, Tsukuba, Japan (11th - 13th Dec, 2024)

13. High Energy Physics in the Quantum Era, KEK Tsukuba Campus, Japan (2-4th December, 2024). [Receipt]

12. (Virtual) The Physical Society of Japan 2024 Spring Meeting (18-21st March, 2024).

11. The 5th KMI school (KMI school 2024): Quantum Computing for Particle Physics and Astrophysics, Nagoya University, Nagoya, Japan (5-7th March, 2024).

10. The 18th Kavli Asian Winter School on Strings, Particles and Cosmology, Yukawa Institute for Theoretical Physics (YITP), Kyoto University, Kyoto, Japan (5 - 14th Dec, 2023) [Certificate]

9. KEK Theory Workshop 2023, KEK Theory Center, Tsukuba, Japan (29th Nov - 1st Dec, 2023) [Receipt]

8. 78th Annual Meeting 2023, The Physical Society of Japan, Tohoku University, Sendai, Japan (16 - 19th Sept, 2023) [Meeting Badge]

7. Field and space-time dynamics by discrete method 2023, University of Tsukuba, Tsukuba, Japan (11-14th Sept, 2023)

6. Spring Workshop on Quantum Gravity, RIKEN, Wako, Japan (26 - 28th April, 2023)

5. The 17th Kavli Asian Winter School on Strings, Particles and Cosmology, Institute for Basic Science, Daejeon, South Korea (9 - 18th Jan, 2023) [Invitation Letter]

4. (Virtual) NTU-Kyoto High Energy Physics workshop, National Taiwan University, Taiwan (17 - 20th Dec, 2022)

3. (Virtual) KEK Theory Workshop 2022 on String Theory and Quantum Field Theory, KEK Theory Center, Tsukuba, Japan, (7 - 9th Dec, 2022)

2. (Virtual)Nonperturbative and Numerical Approaches to Quantum Gravity, String Theory, and Holography, ICTS-TIFR, Bangalore, India (22nd Aug - 2nd Sept, 2022)

1. (Virtual) Nonperturbative and Numerical Approaches to Quantum Gravity, String Theory, and Holography, ICTS-TIFR, Bangalore, India (18th Jan - 22nd Jan, 2021)

FELLOWSHIPS/AWARDS

- Royal Society International Exchange Grant — **June 2025**
  Purpose : Research collaboration with Dr. Masanori Hanada at QMUL. — London, UK
  Funding agency : The Royal Society of United Kingdom.

- MEXT Scholarship [Certificate] — **Oct 2022 - Sept 2025**
  Purpose : PhD Research at KEK. — KEK, Japan
  Funding agency : Ministry of Education, Culture, Sports, Science and Technology, Japan

- SHE INSPIRE Fellowship — **Aug 2017 - May 2022**
  Purpose : BS-MS Dual Degree Programme at IISER Mohali. — IISER Mohali, India
  Funding agency : Department of Science and Technology (DST), Govt. of India.

- Merit-cum Means Scholarship — **Aug 2021 - May 2022**
  Purpose : Financial assistance to poor and meritorious students. — IISER Mohali, India
  Funding agency : Ministry of Minority Affairs, India.

MAJOR COURSES UNDERTAKEN

- Quantum field theory.
- Mathematical methods for physicists.
- Superstring Theory.
- Quantum Information Theory.
- Research Ethics

TECHNICAL SKILLS

- **Programming Languages:**: C, C++, Python, Fortran, Wolfram, Shell Scripting(Bash, Zsh)
- **IDEs:**: Visual Studio Code (VS Code), Anaconda, Jupyter, Google Colab, Bash, Command Line Interface
- **Web:** HTML, CSS, WordPress
- **Scientific Writing:** Markdown, LaTeX, BibTeX
- **Numerical Analysis and Visualization**: Mathematica, Gnuplot, Root, Matplotlib
- **Version Control**: Git, GitHub/GitLab, Bitbucket
- **AI & ML Tools**: TensorFlow, PyTorch, Keras, GPT's
- **Quantum Computing:** Qiskit, Cirq, Pennylane, QuTiP, IBM Quantum Lab
- **High-Performance Computing (HPC):** MPI, OpenMP, CUDA, OpenACC, SLURM, PBS, CMake, HDF5, FFTW, Intel MKL, GPGPU Programming, Singularity, Docker

MEMBER

- The Physical Society of Japan (JPS).
- MEXT Scholars Association (MSA)
- International Association of Physics Students (IAPS).

ACADEMIC SERVICES

- **Executive Committee**, KEK Open House 2024, Tsukuba, Japan.
- **Student coordinator**, KEK Theory Workshop 2023, Tsukuba, Japan.
- **Student coordinator**, International Conference on Gravitation and Cosmology, IISER Mohali, Punjab, India.
- **Volunteer, Physics and Astronomy Club (IISER Mohali)** - Organized and managed a range of engaging physics-related activities for undergraduate students.
- **Physics Content Creation Intern**, FreeCo Education Pvt Ltd. [Offer Letter]

REFERENCES

- **Prof. Jun Nishimura**
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  High Energy Accelerator Research Organization (KEK), Japan
  Email: jnishi@post.kek.jp
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- **Dr. Masanori Hanada**
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  INSPIRE HEP profile
  Personal Website

- **Dr. Anosh Joseph**
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